RESEARCH NOTE · VOL.1

Is AI Power Demand Real — and Has the Market Already Priced It In?

AI 전력 수요는 진짜인가, 그리고 시장은 이미 알고 있나?

October 2026 · Shinjae Kang

Key Takeaways

1. Demand is real on the numbers: the IEA sees data-centre power use at 2.3x by 2030, and AI data-centre demand grew 50% in 2025 vs 3% for global electricity.
2. It shows up on the meter: Virginia commercial sales are up 73% since 2015, 63 pts ahead of the U.S. (+10%).
3. The market has already moved a lot: four Korean power-equipment names are at 7.7x their start-2023 combined market cap; for 4 of 4, price outran profit.

This note answers two questions with public data. First, is AI power demand real? We test it with IEA projections and metered sales data from the U.S. Energy Information Administration (EIA). Second, has the market already priced it in? We set the market caps (KRX) of four Korean power-equipment names against their reported earnings (DART). Every figure is computed from data collected and validated automatically from official sources, with source and as-of date noted.

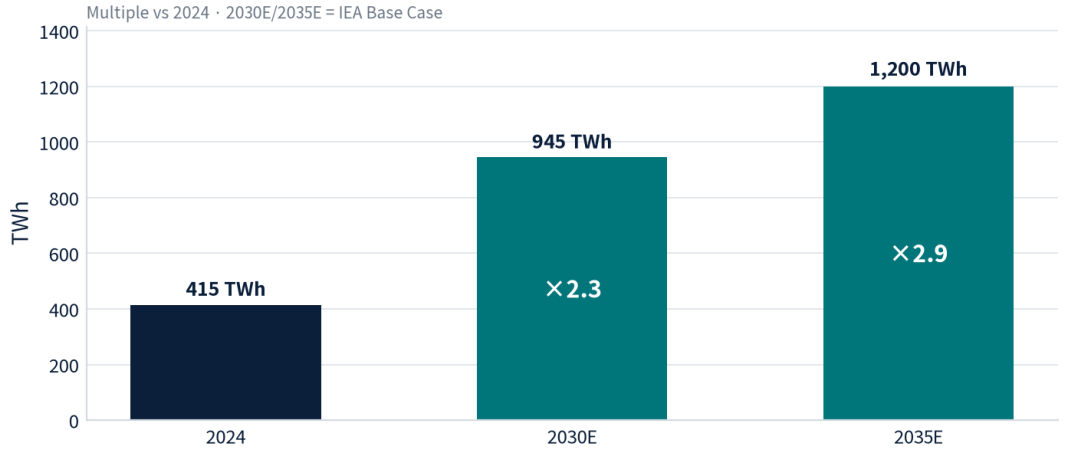
Bottom line. AI power demand shows up in both the outlook (IEA) and metered sales (EIA) — it is real. The market largely knows — the question now is how fast earnings catch up with expectations (author's view). Vol.2 will look at how these expectations feed into the IPO price range of a hypothetical unlisted power-equipment company (Target Co.).

1. Is demand real? — Global outlook

The IEA expects global data-centre electricity use to rise from 415 TWh in 2024 to 945 TWh in 2030 (2.3x) and 1,200 TWh by 2035 (2.9x) in its Base Case. Most of the increase is front-loaded before 2030, which pulls forward grid and transformer investment.

EXHIBIT 1

Data-centre power use set to reach 2.3x by 2030 — 415 TWh (2024) to 1,200 TWh (2035)

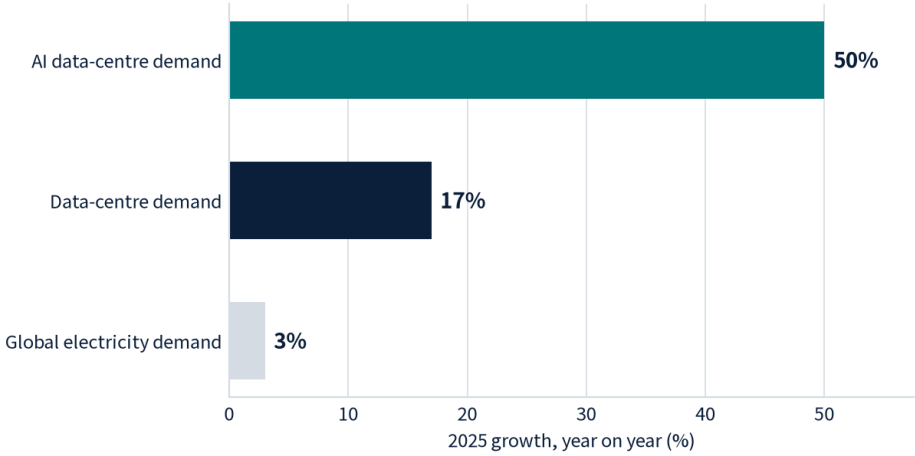


Source: IEA, Energy and AI (Apr 2025) — 2030/2035 are Base Case projections. KSJ Research

Data-centre electricity demand rose 17% in 2025, and AI-focused data centres 50%. Against 3% growth in global demand, AI has become a leading driver of incremental load.

EXHIBIT 2

AI data-centre demand grew 50% in 2025 — 17x the pace of global electricity demand (3%)



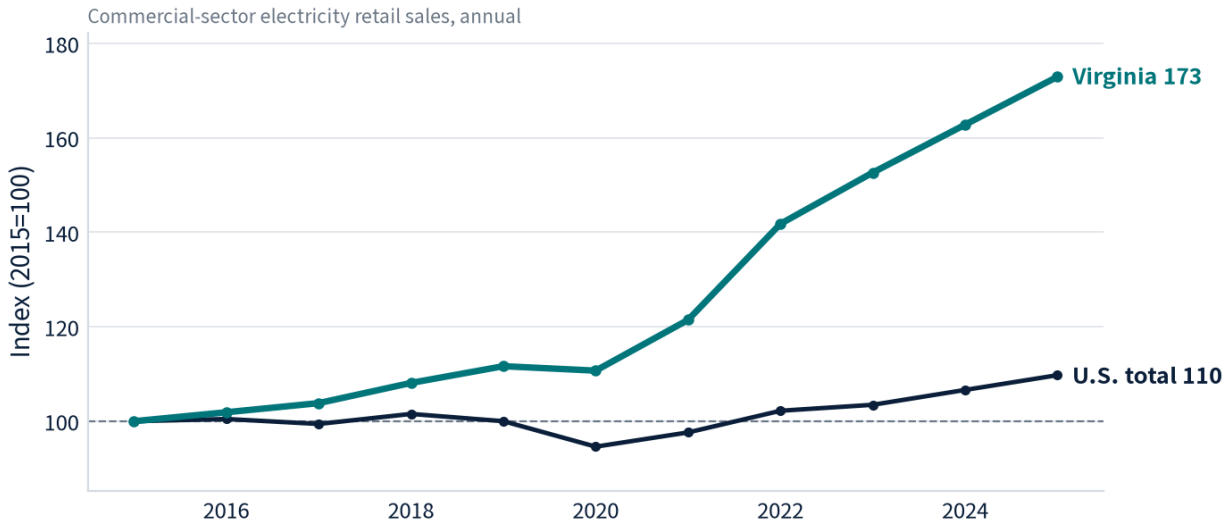
Source: IEA, Key Questions on Energy and AI (Apr 2026). KSJ Research

2. Ground truth — U.S. commercial power sales

Indexed to 2015=100, Virginia stood at 173 and the U.S. at 110 in 2025 (CAGR 5.6% vs 0.9%). Faster growth where data centres cluster suggests AI-related load is already showing up on the meter, not just in forecasts.

EXHIBIT 3

Virginia commercial power sales up 73% since 2015, outpacing the U.S. (+10%) by 63 pts



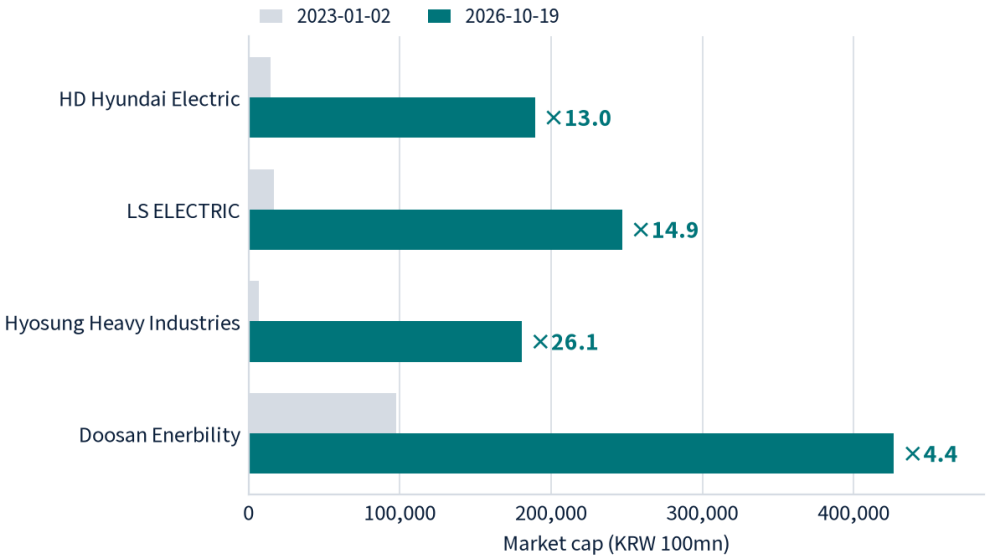
Source: U.S. Energy Information Administration (EIA), electricity retail sales (commercial, annual). KSJ Research

3. Has the market priced it? — Four Korean names

The four names' combined market cap moved from 135,381 (KRW 100mn) on 2023-01-02 to 1,043,284 (KRW 100mn) on 2026-10-19 (7.7x). Hyosung Heavy Industries rose the most (26.1x).

EXHIBIT 4

Combined market cap of four Korean power-equipment names at 7.7x its start-2023 level — Hyosung Heavy Industries leads at 26.1x



Source: Korea Exchange (KRX) Open API, KOSPI daily trading data (market cap). KSJ Research

Table 1. Market-cap multiple = operating-profit multiple × valuation (market cap / OP) multiple

Company	Mkt-cap x	OP x	Mcap/OP (start)	Mcap/OP (latest)	Valuation x
HD Hyundai Electric	13.0x	3.2x	4.6	19.0	**4.1x**
LS ELECTRIC	14.9x	1.3x	5.1	57.9	**11.4x**
Hyosung Heavy Industries	26.1x	2.9x	2.7	24.2	**9.0x**
Doosan Enerbility	4.4x	0.5x	6.6	55.9	**8.4x**

Market cap: first trading day of 2023 → last trading day before the as-of date (KRX). Operating profit: FY2023A → FY2025A consolidated (DART). Valuation = market cap ÷ that year's OP. Source: KRX, DART. KSJ Research

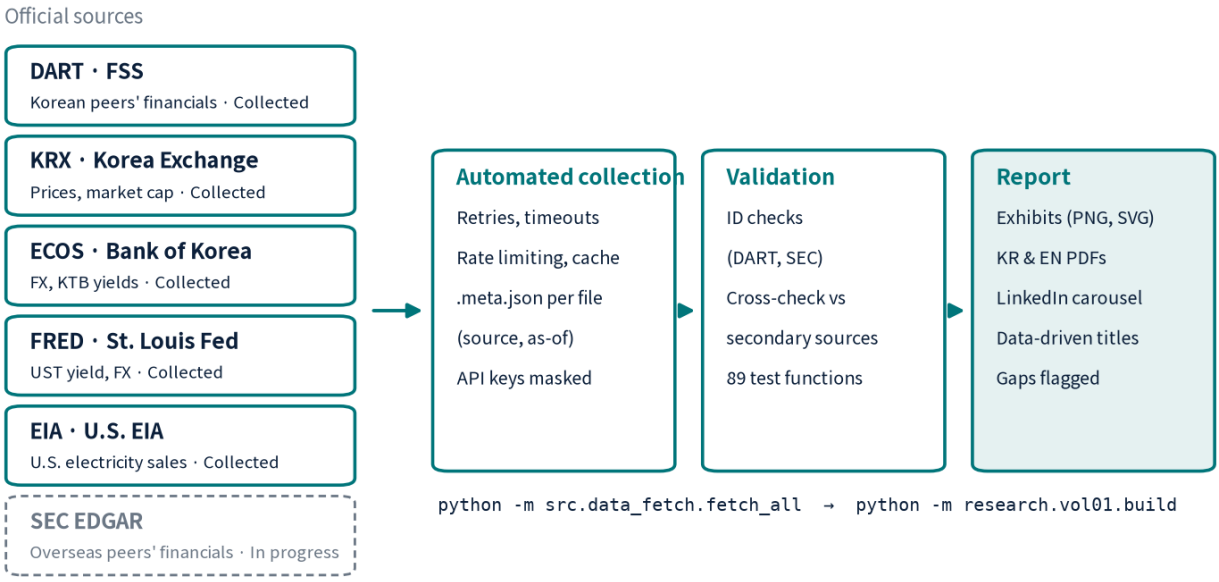
For all 4 names, market cap grew faster than operating profit, expanding valuation multiples. In the author's view, prices are discounting future earnings ahead of reported results.

4. What I built — Data pipeline

Every number in this note's charts, tables and text is computed by code from data files. APIs from 6 official sources are collected through a shared module with retries, timeouts and caching, and every raw file carries a .meta.json with source, as-of date and fetch time. After collection, identifiers (DART corp codes, SEC CIKs) are validated, secondary sources are cross-checked, and units and FX are re-checked; core calculations are covered by 89 test functions. API keys are masked in logs.

EXHIBIT 5

6 official sources → automated collection → validation → report, reproducible with two commands



Source: KSJ Research (src/data_fetch, research/vol01). SEC collection in progress (temporary access block)

Data source status

Source	Use	Status	As of
DART · FSS	Korean peers' financials	**Collected**	2026-10-20
KRX · Korea Exchange	Prices, market cap	**Collected**	2026-10-20
ECOS · Bank of Korea	FX, KTB yields	**Collected**	2026-10-20
FRED · St. Louis Fed	UST yield, FX	**Collected**	2026-10-20
EIA · U.S. EIA	U.S. electricity sales	**Collected**	2026-10-20
SEC EDGAR	Overseas peers' financials	In progress	—

Methodology

- As-of date: 2026-10-20. Market data points: 2023-01-02 (first trading day of 2023) and 2026-10-19 (last trading day before as-of).
- Pipeline: src/data_fetch (shared collection and validation) → research/vol01/fetch_extra.py (extra data) → research/vol01/build.py (exhibits and PDFs).
- Exhibits 1–2: headline figures from public IEA reports, entered in data/manual/iea_energy_ai.csv with source URL and month.
- Exhibit 3: EIA electricity/retail-sales, commercial sector, annual. Index = year's sales ÷ 2015 × 100. Virginia was chosen to test the common view that it is a data-centre hub.
- Exhibit 4 / Table 1: market cap (MKTCAP, KRW → KRW 100mn) from KRX daily trading data; operating profit from DART annual reports (consolidated).
- Limitations: data-centre load may be classified as industrial in some states, so commercial sales show only part of it. Market-cap changes include share-count changes. The market-cap window (from Jan 2023) and the earnings window (FY2023A–FY2025A) do not match exactly.
- SEC (overseas peers' financials) is in progress due to a temporary access block and is not used in this note.

Disclosures

- This is a personal portfolio for education and research. It is not investment advice or a recommendation to buy or sell any security.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Investors are solely responsible for their decisions.
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